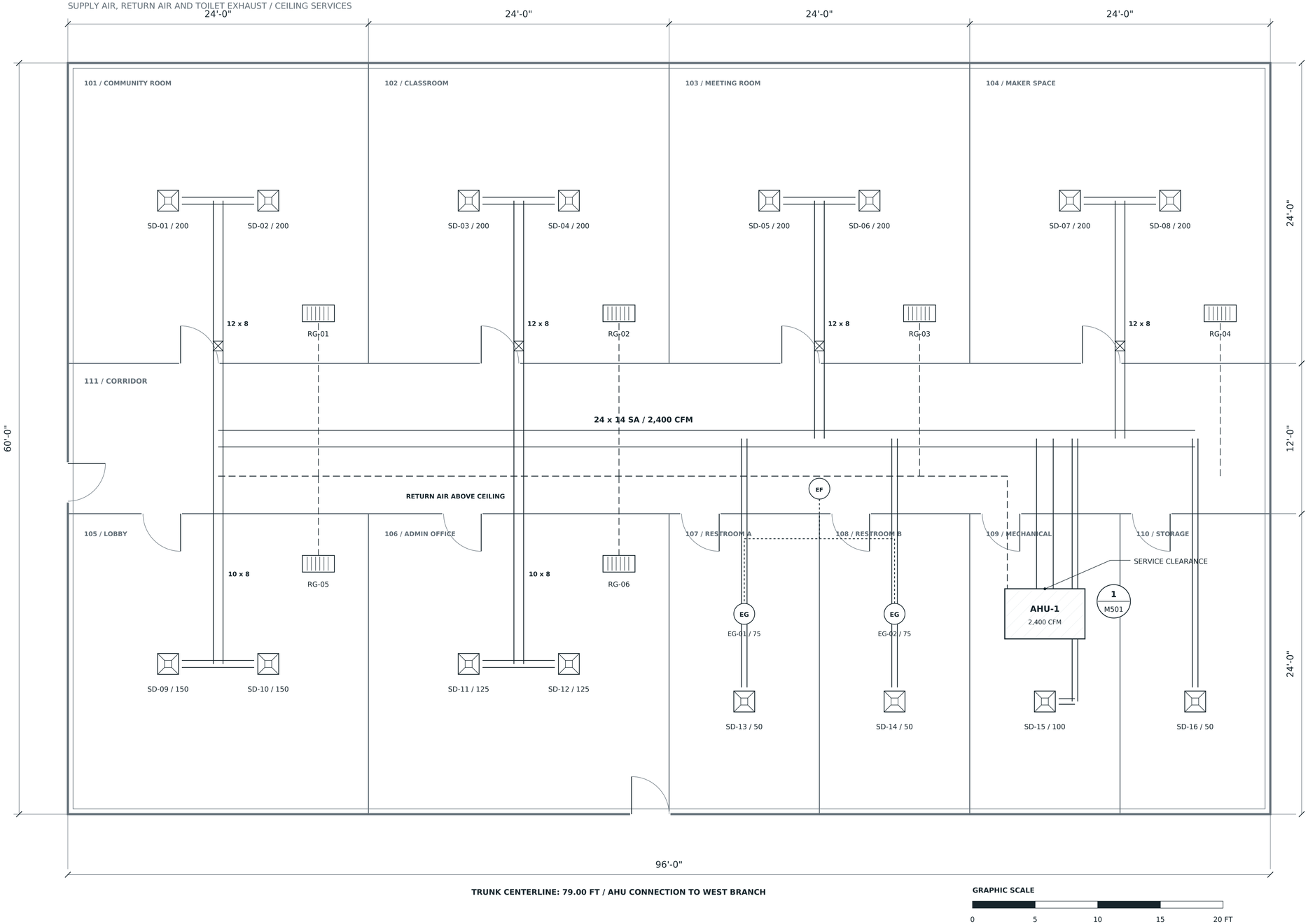
1 / HVAC FLOOR PLAN

SUPPLY AIR, RETURN AIR AND TOILET EXHAUST / CEILING SERVICES



TRUNK CENTERLINE: 79.00 FT / AHU CONNECTION TO WEST BRANCH

GRAPHIC SCALE

0 5 10 15 20 FT

MECHANICAL KEYNOTES

01 Coordinate overhead services with ceiling grid and structure.

02 Provide balancing damper at each supply branch; valve-like symbol denotes damper.

03 Return grilles use ceiling plenum and a ducted return connection at AHU-1.

04 Toilet exhaust manifold connects to roof-mounted EF-1. Provide accessible backdraft damper.

05 AHU-1 is drawn with a nominal service zone. See linked detail 1/M501.

06 All values are invented demonstration data. They are not a verified equipment selection.

PLAN SYMBOLS

- SUPPLY DIFFUSER
- RETURN GRILLE
- EXHAUST TERMINAL
- SUPPLY DUCT
- RETURN PATH
- EXHAUST DUCT

COUNTING EXERCISE

Plan: 16 supply diffusers, 6 return grilles, 2 exhaust terminals.

Legend adds one example of each terminal symbol. Restrict count selection to the plan.

Calibrate using the 20 FT graphic scale: exactly 360 PDF points.

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PROJECT

WILLOW CREEK

COMMUNITY CENTER

WC-26-014

ONE-STORY / 5,760 SF

DRAWING TITLE

HVAC DISTRIBUTION PLAN

DISCIPLINE

MECHANICAL / HVAC

ISSUE RECORD

REVISION A

2026-09-15

Demonstration issue

Drawn: PP / Checked: QA

SHEET DATA

Scale: 1/4 IN = 1 FT

Original: 36 x 24 IN

PDF page 1

Units: feet unless noted

USE NOTICE

FICTIONAL PROJECT

DEMONSTRATION ONLY

NOT FOR CONSTRUCTION

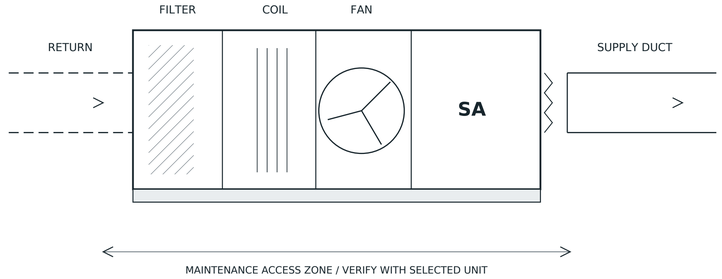
No engineering design, code review or permit approval is implied.

M101

REV A

1 AIR HANDLER CONNECTION DETAIL

DIAGRAM / NOT TO SCALE

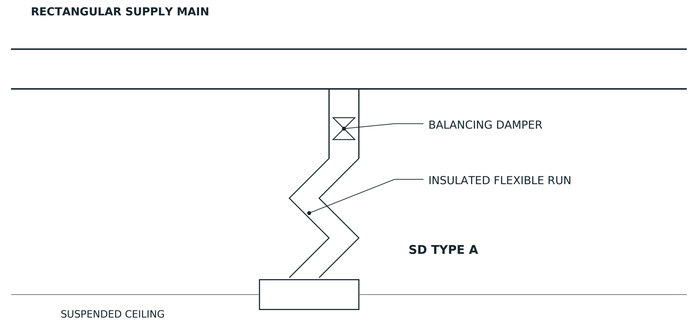


Flexible duct connector at discharge. Vibration pads below unit.

Condensate trap and cleanout at coil drain; route to indirect receptor.

2 SUPPLY DIFFUSER / BRANCH CONNECTION

DIAGRAM / NOT TO SCALE



Keep flex bends smooth. Final neck size by demonstration schedule below.

3 TERMINAL / EQUIPMENT SCHEDULE

SCHEDULE / NOT TO SCALE

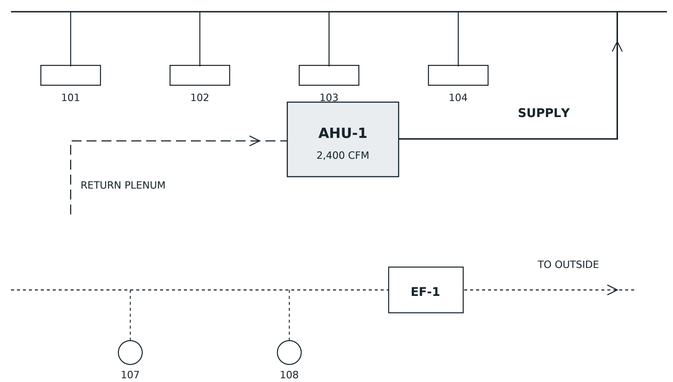
TYPE	DESCRIPTION	QTY	CFM/EA	NECK	NOTES
SD-A	SUPPLY	8	200	8 IN	ROOMS 101-104
SD-B	SUPPLY	2	150	8 IN	LOBBY 105
SD-C	SUPPLY	2	125	6 IN	ADMIN 106
SD-D	SUPPLY	3	50	6 IN	107, 108, 110
SD-E	SUPPLY	1	100	6 IN	MECHANICAL 109
RG-A	RETURN	6	VARIES	12 x 6	CEILING GRILLE
EG-A	EXHAUST	2	75	6 IN	RESTROOMS
AHU-1	AIR HANDLER	1	2,400	N/A	SEE DETAIL 1
EF-1	EXHAUST FAN	1	150	N/A	ROOF MOUNTED

SUPPLY TOTAL: 2,400 CFM / 16 DIFFUSERS / 9 DEMO SCHEDULE ROWS

CFM = cubic feet per minute. Table quantities refer to plan symbols only.

4 DUCT ROUTING / SYSTEM SCHEMATIC

DIAGRAM / NOT TO SCALE



DETAIL NOTES

Details are diagrams, not scaled equipment fabrication drawings.

All equipment tags, terminal quantities and flows are coordinated with M101.

The 79 FT trunk example follows AHU-1 at (78,17) to (78,30) then (12,30).

Branch dimensions denote nominal duct sections in inches.

Use PDF selection, markups, hyperlinks and schedule extraction to demonstrate workflows.

CALIBRATION REMINDER

M101 floor plan is 1/4 IN = 1 FT on a 36 x 24 IN page.

M501 details and schematics are NTS; do not use plan calibration to measure them.

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DRAWING TITLE

HVAC DETAILS AND SCHEDULES

DISCIPLINE

MECHANICAL / HVAC

ISSUE RECORD

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Drawn: PP / Checked: QA

SHEET DATA

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PDF page 2

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M501

REV A